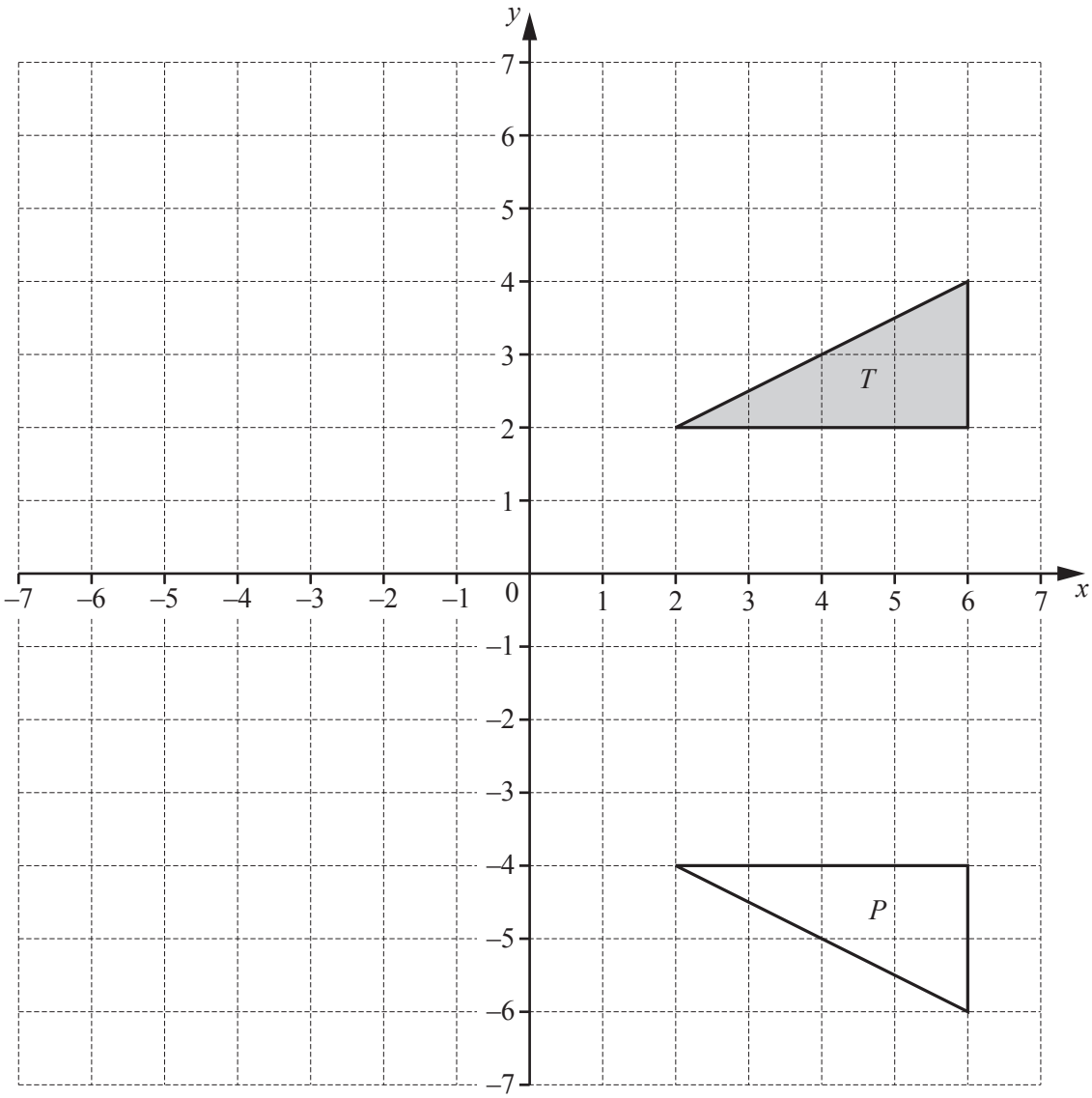
0580_w18_qp_43_Q1

1 (a)



(i) Describe fully the **single** transformation that maps triangle T onto triangle P.

.....

..... [2]

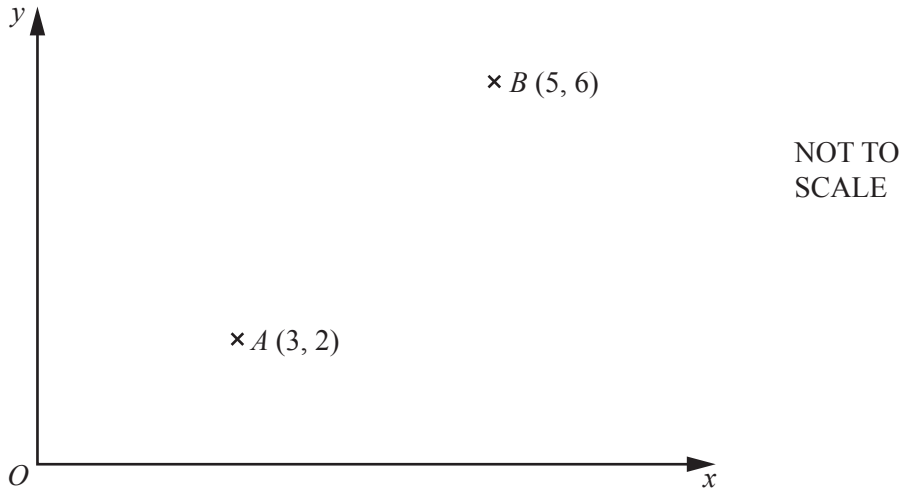
(ii) Translate triangle T by the vector $\begin{pmatrix} -2 \\ -5 \end{pmatrix}$. [2]

(iii) Rotate triangle T through 90° anticlockwise about (0, 0). [2]

(iv) Enlarge triangle T by scale factor $-\frac{1}{2}$ with centre (0, 0). [2]

0580_w18_qp_43_Q1

(b)



(i) Find the column vector $\overrightarrow{AB}$.

$\overrightarrow{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [1]

(ii) Find $|\overrightarrow{AB}|$.

$|\overrightarrow{AB}| =$ ..... [2]

(iii) B is the mid-point of the line AC.

Find the co-ordinates of C.

(..... , ...) [2]

(iv) Find the equation of the straight line that passes through A and B.

..... [3]

(v) The straight line that passes through A and B cuts the y-axis at D.

Write down the co-ordinates of D.

(..... , ...) [1]